Mini Project 1: 5G-NR Interleaver

Date of submission: 15th March 2026

Note:

- You may discuss implementation ideas with classmates, but the final submission must reflect your own understanding.
- Identical or nearly identical submissions will get zero marks.
- Write clean, readable code with proper comments rather than complex solutions.
- You may be asked to present or explain your code individually. During the presentation, you should be able to i) explain the logic and flow of your program, and ii) justify key design choices.
- In order to consider your implementation valid, the latency must be less than 260μs, meeting 5G timing requirements.

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1 5G-NR Bit Interleaving for PDSCH [clause 5.4.2.2 of TS38.212]

The input to the bit interleaving procedure is a binary sequence of E bits, $\boldsymbol{e} = [e_0, \cdots, e_{E-1}]$. As shown in Figure 1, the interleaving is performed as follows,

- The input sequence of E bits is arranged into a matrix with Q_m (modulation order) rows and $\frac{E}{Q_m}$ columns. The number of rows Q_m can take only the values, $Q_m \in \{2, 4, 6, 8\}$.
- The input bits are written into the matrix row-wise, filling the first row from left to right, continuing with the subsequent rows in the same manner until all bits are written, as shown in Figure 1.
- The output sequence is obtained by reading the matrix column-wise, reading the first column from top to bottom, continuing with the remaining columns, as shown in Figure 1.
- The 3GPP procedure for the bit interleaving is given in Algorithm 1 below.

Algorithm 1: 3GPP procedure for bit interleaving

```
// bit interleaving
for j = 0 to E/Q_m − 1 do
    for i = 0 to Q_m − 1 do
        f_{i+j×Q_m} = e_{i×E/Q_m+j}
    end
end
```

1.1 Example

For $E = 16$ and $Q_m = 4$, we get $\frac{E}{Q_m} = 4$ and $\boldsymbol{e} = [e_0, e_1, e_2, \ldots, e_{15}]$.

Input sequence: writing row wise.

$$\begin{matrix} e_0 & e_1 & e_2 & e_3 \\ e_4 & e_5 & e_6 & e_7 \\ e_8 & e_9 & e_{10} & e_{11} \\ e_{12} & e_{13} & e_{14} & e_{15} \end{matrix}$$

Output sequence: $\hat{\boldsymbol{e}} = [e_0, e_4, e_8, e_{12}, e_1, e_5, e_9, e_{13}, e_2, e_6, e_{10}, e_{14}, e_3, e_7, e_{11}, e_{15}]$, i.e., reading column wise.

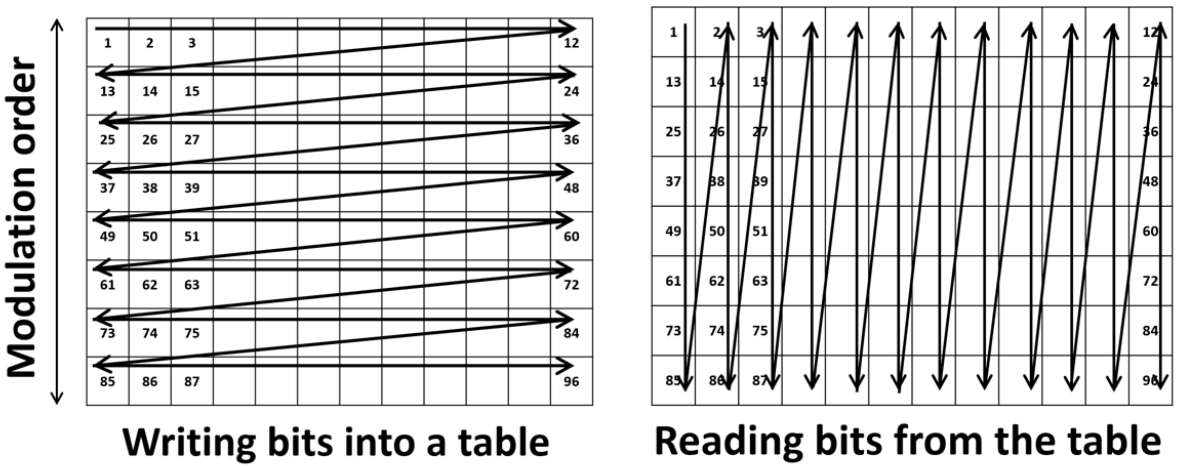

Figure 1: Bit interleaving procedure.

1.2 Problem statement

Realize the above Bit Interleaving procedure into an IP with the name `Interleaver`, using Vivado HLS, with input and output ports being AXI-Stream interfaces. The IP should have

- Input port `inData` with data of width 96 bits and a last signal.
- Output port `outData` with data of width 96 bits and a last signal.
- Control port `cnData` with data of width 128 bits which carries configuration parameters. The parameter E is mapped to bit positions $[14, 0]$, and the parameter Q_m is mapped to bit positions $[18, 15]$.

- There are 14 test cases that must be successfully passed. The corresponding values of E and Q_m are provided in the excel sheet uploaded under **Mini project 1** in LMS. The test vector files required for verification are also available in the same section of the LMS.
- Optimize your implementation as much as you can.

1.3 Submission Requirements

1. Prepare a PowerPoint presentation (not more than 7 slides) that includes the following:
   - Answer why interleaving is important for 5G-NR PDSCH (not more than 1 slide).
   - Why 96 bits input and output data bus width may be a good design choice.
   - **Implementation Overview**: Clearly explain your design approach and implementation methodology.
   - **C/C++ Simulation (CSIM)**: Include a screenshot showing that all test cases pass successfully in a single run.
   - **Synthesis Results**: Include the synthesis report (resource utilization, latency, etc.).
   - **RTL/Cosimulation**: Include a screenshot showing that RTL/Cosim has passed successfully.
   - **Waveform**: Include a waveform screenshot of your implementation.
2. **Files to Submit**: Submit a ZIP file containing, PPT, header file (.h), source code (.cpp), and testbench (.cpp). Name of the ZIP file should be in the format `GroupNumber_RollNumbers.zip`. Any one member of the group can submit the files. Make sure we should be able to replicate the results given in the PPT using the above files.